UNIT 5 · TOPIC 3

Fatigue, Rest Pauses and Comfort

Syllabus topic: Fatigue and comfort

Fatigue

Fatigue = a reversible fall in work capacity caused by sustained effort.

Type	Cause
Muscular	Lactate accumulation, glycogen depletion
Central	Monotony, vigilance decline

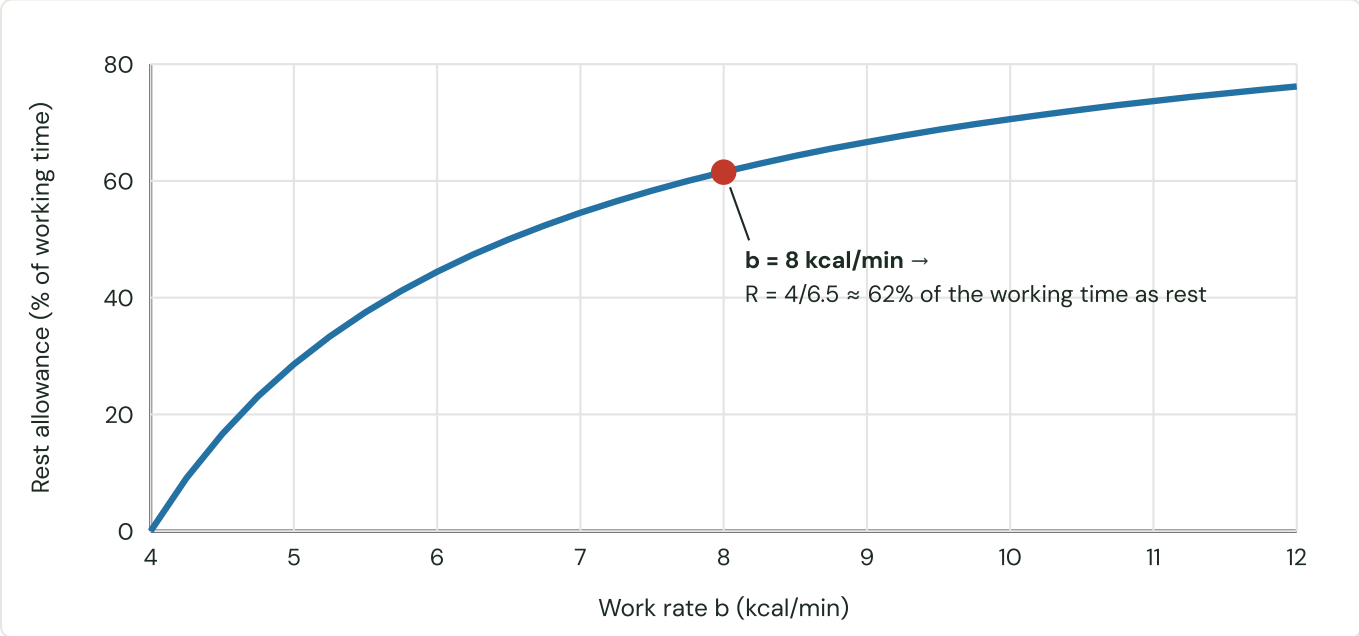
- **Symptoms:** falling output and accuracy, rising errors and accidents, subjective tiredness.
- **Assessment:** physiological (HR recovery, VO_2), performance tests and subjective scales — the **Overall Discomfort Rating (ODR**, 10-point visual analogue) and the **Body Part Discomfort Score (BPDS**, Corlett–Bishop body map).

Rest pauses — Murrell's formula

$$R = T(b - s)/(b - 1.5)$$

R = rest (min) in total time T; b = actual work rate (kcal/min); s = standard sustainable rate (~4–5 kcal/min); 1.5 kcal/min = resting metabolism.

Example. b = 8, s = 4 kcal/min → R/T = (8 - 4)/(8 - 1.5) = 4/6.5 = **61.5%** of the working time as rest.



Murrell's rest allowance vs work rate, s ≈ 4 kcal/min (redrawn from the source figure)

- Frequent SHORT pauses beat a few long ones.
- Task rotation spreads the load across muscle groups.

Thermal and environmental comfort (Liljedahl pp. 204–207)

- Comfort and bearable zones are defined for temperature, humidity, ventilation and thermal radiation (Table 9-1, p. 205).
- **Thermal comfort (ASHRAE):** most people are comfortable at effective temperature **ET* 24–27 °C** (Fig. 9-2, p. 206) — combinations of dry- and wet-bulb temperatures inside the comfort zone.

Dust — the case for pressurized cabs (p. 204)

Item	Value
Combine-cab dust guideline (Netherlands)	15 mg/m³
Tillage dust, peak / mean	577 / 146 mg/m³
Breathing-zone dust in a cab pressurized to 50 Pa with fine filters	24.7 mg/m³ (~6× lower)

India — low-cost equivalents: shading/umbrella, scheduling work in cooler hours, hydration.

Practice MCQs

Q25. Murrell's rest allowance is R =

(a) T(b-s)/(b-1.5) (b) T·b/s
(c) (b+s)/T (d) T-b-s

Q26. In Murrell's formula, 1.5 kcal/min represents:

(a) Standard work rate (b) Resting metabolic rate
(c) Maximum rate (d) Sleep rate

Q27. For b = 8 and s = 4 kcal/min, rest fraction ≈

(a) 20% (b) 40%
(c) 62% (d) 80%

Q28. Better fatigue management favours:

(a) One long break (b) Frequent short pauses and task rotation
(c) No breaks (d) Faster pace

Q29. The 10-point subjective scale for whole-body discomfort is the:

(a) BIS scale (b) ODR (overall discomfort rating)
(c) pH scale (d) Beaufort scale

Q30. Localised discomfort is mapped using the:

(a) Body part discomfort score (Corlett–Bishop) (b) Nebraska chart
(c) MWD (d) NDVI

Q31. ASHRAE effective-temperature comfort zone (Liljedahl p. 206) is about:

(a) 15–18 °C (b) 24–27 °C
(c) 30–35 °C (d) 37–40 °C

Q32. The dust-concentration guideline for combine cabs cited (p. 204) is:

(a) 1 mg/m³ (b) 15 mg/m³
(c) 150 mg/m³ (d) 500 mg/m³

Q33. A tractor cab pressurized to 50 Pa with fine filters reduced mean breathing-zone dust from 146 to about:

(a) 100 (b) 75

(c) 24.7 (d) 2 mg/m³

Q34. Effect of rising fatigue on work is:

(a) More output (b) Falling output & accuracy, more errors and accidents

(c) No change (d) Better vigilance

Answers

Q25 — (a) T(b-s)/(b-1.5)

Q26 — (b) Resting metabolic rate

Q27 — (c) 62% — *R/T = (8 - 4)/(8 - 1.5) = 4/6.5 = 61.5%.*

Q28 — (b) Frequent short pauses and task rotation

Q29 — (b) ODR (overall discomfort rating)

Q30 — (a) Body part discomfort score (Corlett–Bishop)

Q31 — (b) 24–27 °C — *pp. 204–206 — Table 9-1 comfort zones; ET* 24–27 °C.*

Q32 — (b) 15 mg/m³ — *p. 204 — 15 mg/m³ guideline (Netherlands).*

Q33 — (c) 24.7 — *p. 204 — 577/146 → 24.7 mg/m³ with a 50-Pa pressurized cab.*

Q34 — (b) Falling output & accuracy, more errors and accidents

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 9 "Human Factors in Tractor Design", pp. 203–246; CIAE Bhopal / AICRP-Ergonomics literature (Gite & Yadav) for Indian physiological norms; SAE J833; ASHRAE comfort work; NPTEL ergonomics lectures.